

WEEK 11

# Week 11: Cluster Monitoring - Lab Guide

HPC & Cluster Computing

**Objective**: Deploy a complete monitoring stack. **Goal**: Visualize the real-time CPU and RAM usage of your cluster.

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## 1. Prerequisites

**Note:** This lab assumes your Virtual Cluster (configured in previous weeks) is **up and running**. We are adding services to the existing live system.

- [ ] **Master Node**: Will host Prometheus and Grafana.
- [ ] **Compute Nodes**: Will host Node Exporter.
- [ ] **Networking**: Port 9090 (Prometheus) and 3000 (Grafana) must be open.

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## 2. Deploy Node Exporter (Compute Nodes)

We need to start generating metrics.

### 2.1 Update Warewulf Image

- [ ] Enter the container: `bash sudo wwctl container exec rocky-9 /bin/bash`
- [ ] Download and Install (Inside Container): ```bash # Download wget https://github.com/prometheus/node_exporter/releases/download/v1.7.0/node_exporter-1.7.0.linux-amd64.tar.gz tar xvf node_exporter-*.tar.gz cp node_exporter-*.linux-amd64/node_exporter /usr/local/bin/

# Create Systemd Service

cat < /etc/systemd/system/node_exporter.service [Unit] Description=Node Exporter After=network.target

[Service] User=root ExecStart=/usr/local/bin/node_exporter

[Install] WantedBy=multi-user.target EOF

# Enable

systemctl enable node_exporter `- [ ] Exit and Rebuild:` bash exit sudo wwctl container build rocky-9 sudo wwctl power cycle node[01-04] ```

### 2.2 Verify

- [ ] Check if a node is reporting: `bash curl http://node01:9100/metrics | head` *Did you see lines starting with* `node_`*?*

> *"📝 Question 1: Look at the output of the curl command. What is the value of* `node_context_switches_total`*? What does this number imply about the OS kernel?"*

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## 3. Deploy Prometheus (Ma

- [ ] Download and install `bash wget https://github.com/prometheus/prometheus/releases/download/v2.45.0/prometheus-2.45.0.linux-amd64.tar.gz tar xvf prometheus-*.tar.gz sudo mv prometheus-*.linux-amd64 /opt/prometheus`

### 3.2 Configure Targets

- [ ] Edit `/opt/prometheus/prometheus.yml`: ```yaml scrape_configs:
  - job_name: 'cluster' static_configs:
    - targets: ['node01:9100', 'node02:9100', 'node03:9100', 'node04:9100'] ```

> *"📨 Question 2: Why do we specify port `9100` for the targets, but access Prometheus itself on port `9090`?"*

### 3.3 Start Service

- [ ] Run Prometheus (Background): `bash /opt/prometheus/prometheus --config.file=/opt/prometheus/prometheus.yml &`
- [ ] **Verification**: Open Browser to `http://<master-ip>:9090/targets`. Are all endpoints **UP**?

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## 4. Deploy Grafana (Master)

### 4.1 Install

- [ ] Install RPM: `bash sudo dnf install -y https://dl.grafana.com/oss/release/grafana-10.0.0-1.x86_64.rpm`
- [ ] Start Service: `bash sudo systemctl enable --now grafana-server`

- [ ] Open Browser: `http://<master-ip>:3000`
- [ ] Default Login: `admin` / `admin`

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# 5. Build the Dashboard (Interactive)

## 5.1 Add Data Source

- [ ] Click **Connections** → **Data Sources** → **Add new**.
- [ ] Select **Prometheus**.
- [ ] URL: `http://localhost:9090`
- [ ] Click **Save & Test**. *Did it say "Successfully queried"?*

## 5.2 Create Dashboard

- [ ] Click **Dashboards** → **New**.
- [ ] **Add Visualization**.
- [ ] Select **Prometheus** as source.
- [ ] Query: `node_memory_MemTotal_bytes`
- [ ] Click **Run Queries**. *Result*: You should see a graph of memory usage for all your nodes!

> *"📝 Question 3: Run a stress test on `node01` (e.g., `stress --cpu 4`). Watch the Grafana dashboard. How long is the delay between starting the stress test and seeing the spike on the graph? Why is there a delay?"*

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# 6. Submission

Create a Lab Report PDF containing:

1. **Answers**: Provide your written answers for Questions 1, 2, and 3.
2. **Screenshots**:
   - Prometheus Targets page showing all nodes UP.
   - Your Grafana Dashboard showing live data (with the stress test spike visible).